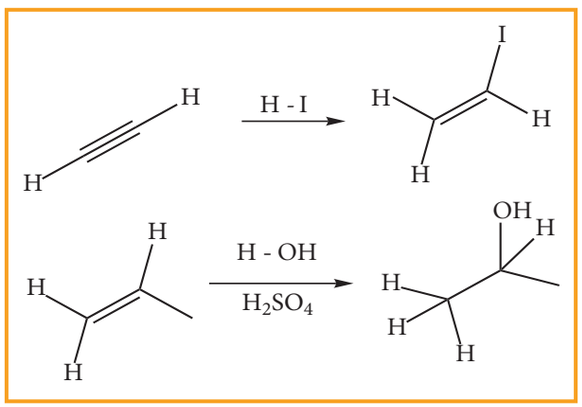
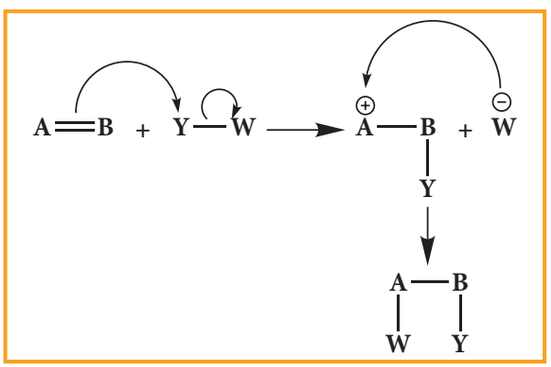
12.2.2 Addition reactions

It is a characteristic reaction of an unsaturated compound (compounds containing C-C localised double or triple bond). In this reaction two molecules combine to give a single product. Like substitution this reaction also can be classified as nucleophilic, electrophilic and freeradical addition reactions depending the type of reagent which initiates the reaction. During the addition reaction the hydridisation of the substrate changes (from $sp^2 \rightarrow sp^3$ in the addition reaction of alkenes or $sp \rightarrow sp^2$ in the addition reaction of alkynes) as only one bond breaks and two new bonds are formed.

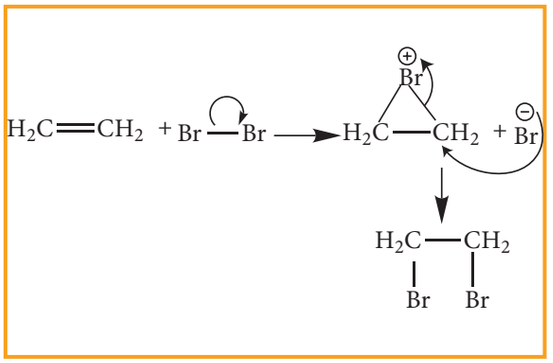


Electrophilic Addition reaction

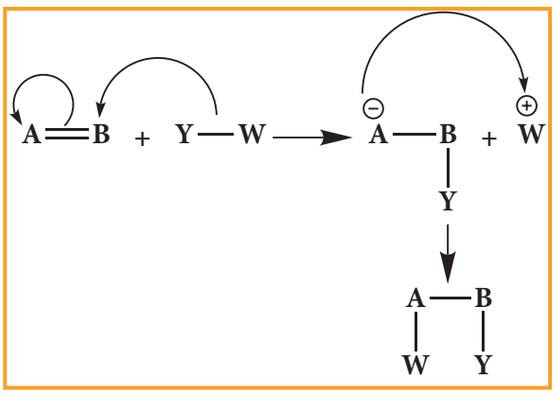
A general electrophilic addition reaction can be represented as below.



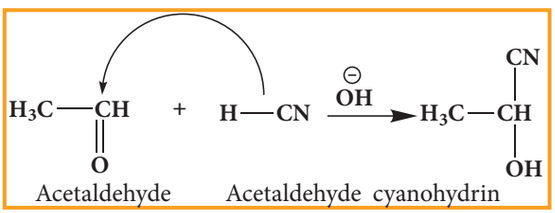
Brominatin of alkene to give bromo alkane is an example for this reaction.



Nucleophilic addition reaction

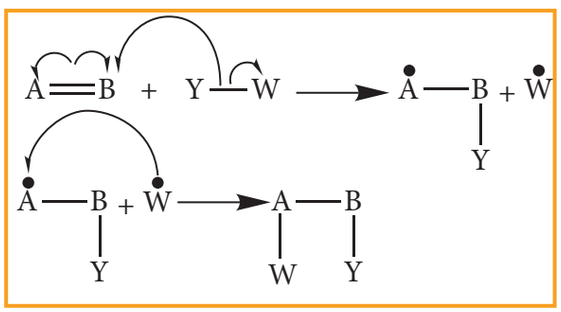


Example: addition of HCN to acetaldehyde



Free radical addition Reaction:

A General freeradical addition reaction can be represented as below.



Unit 12.indd 172 23-12-2021 17:15:09